PEARSON EDEXCEL INTERNATIONAL GCSE

Mathematics A

Unit 2 • Chapter OL-T28 Classified

Cross-session • chapter-organised candidate

1 classified items • 3 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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Contents

Unit 2 • Unit 2 • Chapter OL-T28 • 1 items • 3 marks

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Forming & Solving Equations

Unit 2 • 3 marks • 1 item(s)

November 2025 | 2H Q20 | 3 marks

General geometric formula derivation

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

20 Here is a quadratic equation.

$$ax^2 + 4x + c = 0$$

The solutions of this equation are given by $x = \dfrac{-4 \pm 2\sqrt{39}}{10}$

Find the value of a and the value of c
Show your working clearly.

$a =$ ....

$c =$ ....

(Total for Question 20 is 3 marks)

Worked solution

November 2025 | Question 20 | 3 marks

Family: Derive a general expression or formula • Variant: General geometric formula derivation

Method

For

$ax^2+4x+c=0$

the quadratic formula gives

$x=\frac{-4\pm\sqrt{4^2-4ac}}{2a}$

The given solutions are

$x=\frac{-4\pm 2\sqrt{39}}{10}$

Compare denominators:

$2a=10$

$a=5$

Compare the square root parts:

$\sqrt{16-4ac}=2\sqrt{39}$

Square both sides:

$16-4ac=156$

Use $a=5$:

$16-20c=156$

$-20c=140$

$c=-7$

Answer: $a=5,\ c=-7$

PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION